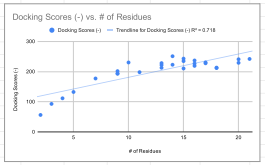
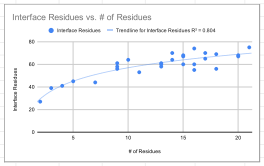
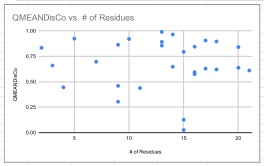
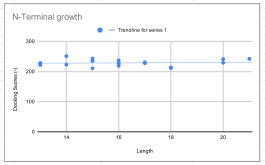
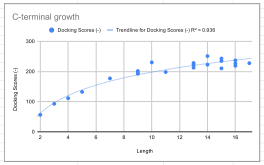
Protein-protein complex	# ΔG (kcal mol-1)	Kd (M) at	# ICs charged-charged	# ICs charged-poler	# ICs charged-apoler	# ICs polar-polar	# ICs polar-apoler	Docking Scores (-)	# of Residues	H-Bond Energy (kJ/mol)	ostatic Energy (kJ	Waals Energy (kJ	abilizing Energy	Interface Residue	alized Energy (kJ	Short Contacts	rophobic Interacti	vdW Pairs	Salt Bridges	Fac. Electrostatic	Infac. Electrostati	QMEANDisCo	RMSD	pLDDT
cypA_GP	-5.5	9.10E-05	2	2	16	0	2	56.177	2	-5.6	0	-49.72	-55.32	27	-2.05	4	0	1738	0	0	0	0.834	0.0238	
cypA_GPI	-5.9	4.40E-05	0	0	8	0	10	92.776	3	25.87	0	-95.46	-121.33	39	-3.11	19	2	3177	0	0	0	0.66	0.0468	
cypA_AGPI	-6.3	2.30E-05	0	0	9	0	10	111.398	4	-37.83	0	-100.03	-137.86	41	-3.36	25	2	3869	0	0	0	0.445	1.134	81.074
cypA_AGPIA	-10.4	2.30E-08	6	3	32	0	17	132.349	5	-15.6	0	-111.23	-126.83	45	-2.82	17	4	4158	0	0	0	0.924	1.6494	96.2
cypA_VHAGPIA	-7.3	4.20E-06	6	8	20	0	8	177.12	7	-12.24	23.72	-118.82	-107.34	44	-2.44	22	5	4915	0	0	3	0.697	1.9538	95.03
cypA_PVHAGPIAP	-8.6	5.10E-07	5	4	19	0	14	193.679	9	-17.47	22.02	-128.44	-123.88	61	-2.03	70	2	7594	0	0	3	0.863	1.542	92.27
cypA_VHAGPIAPG	-4.5	5.40E-04	0	0	4	0	4	192.946	9	-19.28	4.24	-148.18	-163.23	56	-2.91	37	6	6241	0	1	3	0.46	2.6838	91.81
cypA_HPVHAGPIA	-9.7	7.90E-08	5	6	23	5	20	201.49	9	-14.09	21.82	-130.57	-122.85	58	-2.12	39	1	6890	0	0	3	0.304	1.7263	95.25
cypA_HPVHAGPIAP	-5.6	8.30E-05	0	0	6	0	8	230.203	10	-27.58	23.59	-148.64	-152.63	64	-2.38	42	6	8412	0	0	3	0.921	2.4273	95.94
cypA_HPVHAGPIAPG	-8.7	4.00E-07	3	6	21	0	15	198.05	11	-6.21	25.87	-141.21	-121.55	53	-2.29	23	1	6058	0	0	3	0.438	3.2867	95.14
cypA_HPVHAGPIAPGQM	-7	7.30E-06	2	5	16	0	10	212.603	13	-35.7	11.76	123.31	99.37	60	1.66	61	4	8520	0	1	5	0.99	5.2552	90.35
cypA_VHPVHAGPIAPGQ	-9	2.50E-07	7	6	15	0	16	221.661	13	0	48.68	-162.26	-113.58	58	-1.96	31	2	7247	0	0	5	0.856	6.5189	87.97
cypA_WDRVHPVHAGPIA	-9.3	1.60E-07	6	8	27	0	14	228.104	13	-2.74	39.71	-28.51	8.46	61	0.14	52	3	10136	0	3	7	0.892	3.9227	84.66
cypA_HAGPIAPGQMREPR	-7.4	3.90E-06	2	3	17	0	12	222.661	14	-76.28	-9.28	-184.75	-270.3	64	-4.22	57	1	9181	0	3	1	0.648	8.038	91.95
cypA_WDRVHPVHAGPIAP	-10	4.30E-08	2	0	17	0	23	250.913	14	-14.52	23.02	-176.37	-167.87	70	-2.4	89	6	11245	0	2	7	0.965	3.8146	90.35
cypA_DRVHPVHAGPIAPGQ	-7	6.80E-06	1	5	18	0	10	234.716	15	0	35.37	-158.88	-123.51	60	-2.06	48	2	8037	0	0	6	0.794	6.3533	84.34
cypA_VHAGPIAPGQMREPR	-6.6	1.50E-05	7	10	18	2	7	210.302	15	-24.84	-0.57	111.68	86.27	68	1.27	56	0	8003	0	3	4	0.126	4.5664	92.65
cypA_WDRVHPVHAGPIAPG	-8.7	4.40E-07	4	3	19	1	16	242.98	15	-18.74	2.56	7.73	-8.45	67	-0.13	45	1	7457	1	2	6	0.025	6.1076	89.58
cypA_DRVHPVHAGPIAPGQM	-7.5	3.20E-06	5	5	18	0	10	218.915	16	-10.73	37.09	-161.3	-134.94	55	-2.45	40	4	7151	0	1	6	0.845	6.9654	87.1
cypA_PVHAGPIAPGQMREPR	-7	6.80E-06	6	10	22	0	5	226.567	16	-3.34	26.21	-177.67	-154.8	60	-2.58	65	3	8949	0	0	2	0.574	7.3689	90.31
cypA_WDRVHPVHAGPIAPGQ	-8.6	5.30E-07	6	12	32	0	7	236.844	16	-16.01	32.73	-133.32	-116.61	74	-1.58	66	3	9724	0	1	6	0.595	4.9148	85.94
cypA_HPVHAGPIAPGQMREPR	-6.2	2.90E-05	8	9	24	0	1	227.507	17	-24.72	-35.21	-119.46	-179.39	70	-2.56	67	3	9942	2	2	1	0.629	6.4624	91.46
cypA_WDRVHPVHAGPIAPGQM	-10.4	2.40E-08	5	3	33	0	16	230.411	17	-9.79	22.16	-133.28	-120.9	65	-1.86	59	4	8620	0	1	3	0.906	4.0847	82.52
cypA_VHPVHAGPIAPGQMREPR	-6.9	8.50E-06	4	8	28	0	3	213.128	18	-26.94	41.09	-69.26	-55.1	56	-0.98	53	2	8677	0	1	6	0.621	6.2619	92.74
cypA_WDRVHPVHAGPIAPGQMR	-7.6	2.40E-06	7	13	28	0	5	211.69	18	-37.22	39.99	-204.07	-201.3	69	-2.92	55	2	10257	0	2	8	0.896	7.4658	86.41
cypA_DRVHPVHAGPIAPGQMREPR	-7.4	3.90E-06	8	12	17	0	8	240.85	20	-20.19	8.54	-13.68	-25.33	68	-0.37	66	1	7300	2	1	5	0.841	7.1865	86.66
cypA_WDRVHPVHAGPIAPGQMREP	-10.1	3.80E-08	6	9	22	0	18	229.056	20	-26.51	-8.4	-148.68	-183.59	67	-2.74	53	1	8619	0	4	7	0.638	5.9333	86.43
cypA_WDRVHPVHAGPIAPGQMREPR	-9.7	8.30E-08	4	3	28	1	16	241.89	21	-56.62	20.14	-122.79	-159.26	75	-2.12	76	3	10024	3	3	5	0.611	9.4992	88.82

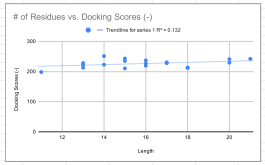


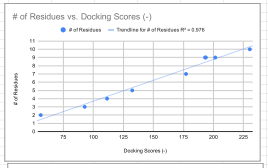


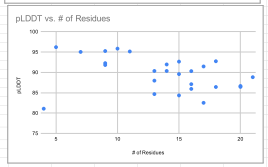


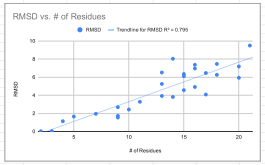












Alanine Scan Using BUDE Alanine Scan (BAlaS)

ΔG (kcal/mol)	Chain	Residue	Amino Acid	ΔΔG (kcal/mol)	Std Dev
-7.313942109	B	2	PRO	2.5334636	0

ΔG (kcal/mol)	Chain	Residue	Amino Acid	ΔΔG (kcal/mol)	Std Dev
-7.781476091	B	2	PRO	1.8164456	0
	B	3	ILE	2.629066	0

ΔG (kcal/mol)	Chain	Residue	Amino Acid	ΔΔG (kcal/mol)	Std Dev
-8.038154 19	B	3	PRO	1.912048	0
	B	4	ILE	2.151054	0

ΔG (kcal/mol)	Chain	Residue	Amino Acid	ΔΔG (kcal/mol)	Std Dev
-14.28290296	B	3	PRO	1.12332382	0
	B	4	ILE	1.0038252	0

ΔG (kcal/mol)	Chain	Residue	Amino Acid	ΔΔG (kcal/mol)	Std Dev
-11.10601131	B	1	HIS	0.4302108	0
	B	2	PRO	1.5776396	0
	B	3	VAL	0.717018	0
	B	4	HIS	1.6969426	0
	B	7	PRO	1.0277258	0
	B	8	ILE	0.1434036	0

ΔG (kcal/mol)	Chain	# Residue	Amino Acid	# ΔΔG (kcal/mol)	# Std Dev
-15.77563883	B	1	TRP	-6.572665	0
	B	2	ASP	0.7887198	0
	B	3	ARG	3.1787798	0
	B	4	VAL	0.0717018	0
	B	5	HIS	0.9799266	0
	B	6	PRO	0.6931174	0
	B	7	VAL	0.6453162	0
	B	8	HIS	0.7409186	0
	B	11	PRO	0.6692168	0
	B	12	ILE	0.5497138	0
	B	14	PRO	0.0717018	0

ΔG (kcal/mol)	Chain	# Residue	Amino Acid	# ΔΔG (kcal/mol)	# Std Dev
-20.2875941	B	1	HIS	0.836521	0
	B	4	PRO	0.7409186	0
	B	5	ILE	0.5258132	0
	B	7	PRO	1.4579366	0
	B	9	GLN	0.358509	0
	B	10	MET	0.6453162	0
	B	11	ARG	0.0239006	0
	B	12	GLU	2.4139606	0
	B	13	PRO	0.4302108	0
	B	14	ARG	0.0478012	0

WDRVHPVHAGPIAPGQMREPR

ΔG (kJ/mol)	Chain	# Residue	Amino Acid	# ΔΔG (kJ/mol)	# Std Dev
-20.72215481	B	3	ARG	0.5019126	0

Protein-protein complex	#	ΔG (kcal mol-1)	Kd (M) at	#	ICs charged-charged	#	ICs charged-polar	#	ICs charged-apolar	#	ICs polar-polar	#	ICs polar-apolar	Docking Scores (-)	# of Residues	H-Bond Energy (kJ/mol)	ostatic Energy (kJ	Waals Energy (kJ	abilizing Energy	Interface Residuals	Residualized Energy (kJ	Short Contacts	rophobic Interacti	vdW Pairs	Salt Bridges	Fav. Electrostatic	Unfav. Electrostati	QMEANDisCo	RMSD	pLDDT
	B		4	VAL			0.0239006		0																					
	B		5	HIS			0.8126204		0																					
	B		6	PRO			0.2151054		0																					
	B		7	VAL			1.4101354		0																					
	B		8	HIS			0.5019126		0																					
	B		11	PRO			1.5774396		0																					
	B		12	ILE			1.2628312		0																					
	B		14	PRO			0.6892168		0																					
	B		16	GLN			1.434036		0																					
	B		17	MET			0.0239006		0																					
	B		18	ARG			0.2629066		0																					
	B		21	ARG			0.1673062		0																					